How Model Transformations Affect Fingerprints in Transformer Key–Value Caches

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Abstract

When a model is copied or modified, its owner needs a practical way to check whether a deployed model is the same model. A model fingerprint is a small hidden signal placed inside the model and later checked during inference. This check becomes unreliable when deployment changes the model through quantization, pruning, weight mixing, or cache eviction. We test a simple way to make the signal tolerate such changes: protect seven bits with a Bose–Chaudhuri–Hocquenghem (BCH) error-correcting code, spread the coded bits across several cache coordinates, and decode them from matched clean and marked runs. We evaluate four language models, three keys per model, and 14 transformations. The signal is recovered perfectly in the native and 8-bit integer (INT8) settings, but 2-bit integer (INT2) quantization usually destroys it. With 30% pruning, recovery ranges from 78.3% to 100% across models; with 50% pruning it ranges from 8.3% to 93.3%. A quantization index modulation (QIM) comparison shows that coding helps under some pruning settings but adds no benefit when the signal is already intact. Simulated cache eviction can preserve detection while making the model much worse at predicting the next token.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Security and privacy** → **Logic and verification**.

Keywords

Key–value cache, model auditing, error-correcting codes, quantization index modulation, pruning, quantization

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1 Introduction

Large language models are reused in many places [4]. A model may be downloaded, fine-tuned, compressed, or served by a different provider [17]. When ownership or unauthorized reuse is disputed, the owner needs evidence that can be checked from the model itself, independent of a file name or a provider's claim [8]. A model fingerprint is a secret, model-specific signal embedded in or induced by the model and recovered through a designated verification procedure [20]. The decoder reports whether the deployed model still carries the enrolled fingerprint.

The same deployment changes can affect the signal and the model in different ways. Weight quantization rounds the values that carry the signal. Projection pruning removes or attenuates selected parameters, including coordinates that may carry the verification signal. Weight mixing can attenuate or interfere with the signal [19]. KV-cache eviction hides part of the inference history [21]. These operations may leave language quality acceptable while making a fingerprint decoder fail. A decoder can also recover the signal from a severely degraded model. We measure both outcomes: fingerprint recovery and the model's prediction quality after each transformation.

Our design adds redundancy. We encode seven bits with a Bose–Chaudhuri–Hocquenghem (BCH) code with parameters (15,7,5) [1, 13], place the 15 coded bits on separate Value coordinates, and use a low-amplitude spread carrier [3, 12]. The code corrects up to two bit errors in each codeword, while the coordinate layout limits the effect of a localized failure [16]. We compare this design with the same code without interleaving, a repetition code, and an uncoded payload using the same number of physical coordinates and the same perturbation energy.

We study three questions. Under the evaluated quantization, mixing, pruning, and eviction settings, when does the coded fingerprint survive or fail? Does interleaved BCH improve recovery under structured pruning errors? Can reliable detection coexist with severe degradation in next-token prediction? We use matched clean and marked runs across four checkpoints, three public-key seeds, and 14 predefined conditions.

2 Method

2.1 Relation to prior audit channels

Error-correcting codes correct structured symbol errors in communication channels. QIM embeds symbols through quantized reconstruction cells [2], while watermarking has been studied in parameters and generated text [7, 17]. We use numerical Value states observed during inference as the audit channel. Cache quantization

systems such as KIVI and KVQuant target memory and throughput [6, 9]. We apply the corresponding numerical transformations to selected Value coordinates as controlled stress tests.

2.2 Decoder and acceptance

For each physical axis, the recorder stores the signed correlation r_i between the marked-minus-clean activation difference and the enrolled carrier direction. The soft decoder evaluates all 128 BCH codewords after deinterleaving and selects the minimum Euclidean score. A document passes when the selected codeword equals the enrolled message and the average signed correlation exceeds 0.5. The two checks keep a strong response on an incorrect message from being counted as ownership evidence. BCH and ordered BCH carry seven information bits in 15 axes; repetition carries seven bits with repeated symbols; the uncoded control carries seven bits plus eight constant padding axes.

2.3 Coded carrier

Let $b \in \{0,1\}^7$ be a message and let $c = \mathrm{BCH}_{15,7,5}(b)$ be its 15-bit codeword. A fixed public-key permutation π interleaves the codeword over 15 selected physical axes. Each bit is mapped to $s_i \in \{-1,+1\}$ and embedded as a low-rank binary phase-shift keying (BPSK) perturbation

$$\Delta W = \alpha \sum_{i=1}^{15} s_i \, u_i v_i^{\top}, \tag{1}$$

where u_i, v_i are orthonormal factors selected from the Value projection and α is scaled to a Frobenius-norm budget of 0.04. The decoder estimates signed correlations on the same axes, deinterleaves them, and applies soft maximum-likelihood BCH decoding. Acceptance requires the target message and a signed correlation above 0.5.

The ordered-BCH control keeps the same code but removes the spatial permutation. The repetition control repeats information bits to the same physical length. The uncoded control uses the same carrier budget with constant padding. All controls use the same number of physical axes and the same perturbation norm.

2.4 Transformations and metrics

We evaluate native cache states; selected Value-head INT8, INT4, and INT2 quantization; interpolation with the clean weights at mixing factors 0.75, 0.5, 0.25, and 0; selected-projection pruning at 10%, 30%, and 50%; and causal random-priority eviction retaining 75%, 50%, and 25% of tokens. For each condition, the clean and marked routes use the same documents and codec state.

True-positive rate (TPR) is the fraction of marked documents decoded to the enrolled message. False-positive rate (FPR) is measured on clean documents. We also report payload bit error rate (BER), paired document-level bootstrap intervals for coded-minus-uncoded TPR, and the change in log perplexity (PPL) relative to the native clean route.

3 Experimental Setup

The matrix contains Qwen3-0.6B [15], Qwen2.5-0.5B [14], Gemma3-1B-IT [5], and Llama3.2-1B [11]; B denotes billions of parameters and IT denotes instruction-tuned. Inputs are tokenized with the Transformers library [18] from WikiText-2 validation material [10]. Each model uses seeds 2027, 2028, and 2029, 12 selection documents, 16 calibration documents, and 20 paired positive/null test documents. Four payload methods and 14 transformations give 672 conditions and 26,880 document events. The QIM control uses the same selected axes and a signed scalar quantizer whose step is calibrated only from clean selection activations. We use two major open-source models, Qwen3 and Llama3.2, as matched experimental subjects. For each model, seed 2027, the selected axes, payload energy, calibration data, offsets, and 14 transformations are kept unchanged; only the model is changed. We compare BCH and constant-padded uncoded payloads at offsets 0 and 0.25. This design isolates coding effects while checking whether the QIM control transfers across model families.

3.1 SFT and LoRA adaptation

To measure persistence after downstream adaptation, we inserted the fingerprint before training and evaluated checkpoints at epochs 1, 3, 5, and 7 on the same frozen prompt-only audit split. The adaptation data combine Dolly-derived instruction data with a domain language-modeling corpus. Runs use bfloat16 and gradient checkpointing. Full-parameter SFT uses a learning rate of 2×10^{-5}, per-device batch size 1, and 16-step gradient accumulation. LoRA uses a learning rate of 2×10^{-4}, rank 16, alpha 32, dropout 0.05, and query, key, value, and output projection targets, with batch size 2 and 8-step accumulation. Clean and marked routes use matched prompts, and we report TPR, FPR, payload BER, and perplexity at each checkpoint.

3.2 Reproducibility and statistical unit

We treat each document event as one inference unit. Repeated decoder calls do not add observations. Each condition records 20 marked and 20 matched clean events per key, including decoded message, physical and payload BER, signed correlation, retained-token count, and next-token negative log likelihood. Means across the three keys are descriptive because the keys share one checkpoint. Paired bootstrap intervals resample document indices jointly for coded and uncoded routes.

4 Results

4.1 Lifecycle response across transformations

Figure 1 visualizes the severity trajectories, while Table 1 reports all 14 BCH-interleaved conditions for all four checkpoints.

Native and INT8 conditions retain 100% TPR for all checkpoints. INT4 also retains 100% except on Gemma3, where TPR falls to 76.7%; INT2 reduces TPR to 5% on Qwen3 and zero on the other checkpoints. Under weight mixing, TPR remains 100% at a marked fraction of 0.75, falls to 28.3–61.7% at 0.5, and reaches zero at 0.25 and 0. Table 4 determines whether these detection losses arise from bit errors or from the acceptance threshold.

Projection pruning produces the strongest checkpoint-dependent separation. At 30% pruning, TPR ranges from 78.3% to 100%; at 50%, it ranges from 8.3% to 93.3%. Retaining 75%, 50%, or 25% of visible tokens leaves TPR at 100% for the BCH-interleaved payload. The prediction cost of that visibility intervention is evaluated separately in Table 4.

Table 1: Complete BCH-interleaved detection matrix. TPR is the percentage of 20 marked documents averaged over three carrier seeds; FPR is zero in every listed BCH-interleaved row.

Transformation	Qwen3	Qwen2.5	Gemma3	Llama3.2
Native	100.0	100.0	100.0	100.0
INT8	100.0	100.0	100.0	100.0
INT4	100.0	100.0	76.7	100.0
INT2	5.0	0.0	0.0	0.0
Mix 0.75	100.0	100.0	100.0	100.0
Mix 0.50	61.7	28.3	30.0	28.3
Mix 0.25	0.0	0.0	0.0	0.0
Mix 0	0.0	0.0	0.0	0.0
Prune 10%	100.0	100.0	100.0	100.0
Prune 30%	78.3	100.0	83.3	100.0
Prune 50%	43.3	93.3	8.3	93.3
Keep 75%	100.0	100.0	100.0	100.0
Keep 50%	100.0	100.0	100.0	100.0
Keep 25%	100.0	100.0	100.0	100.0

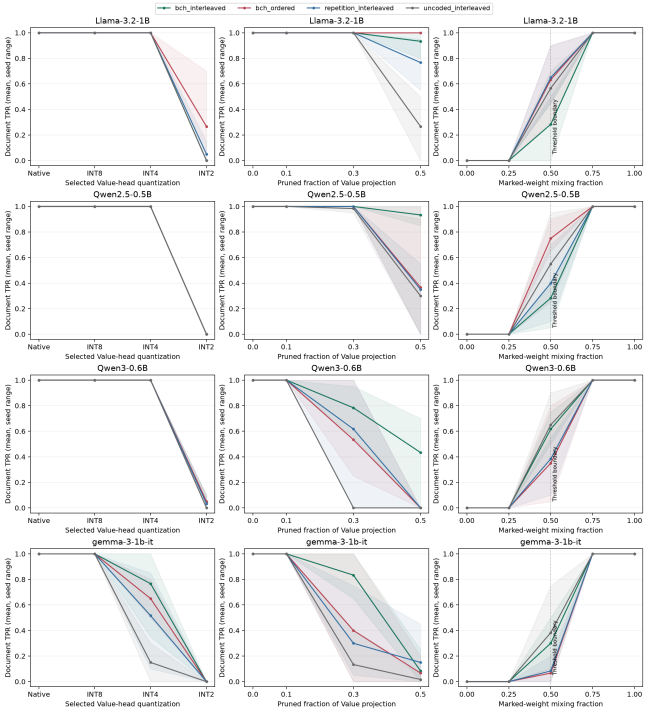


Figure 1: Lifecycle curves for the four model families. Columns show Value-head cache quantization, projection pruning, and marked-weight mixing. Markers show the three carrier seeds and lines show their means.

4.2 Complete matched QIM control

Tables 2 and 3 contain every tested transformation at offsets 0 and 0.25. They jointly report detection, the paired BCH-minus-uncoded contrast, payload BER, and prediction loss for the Qwen3 and Llama-3.2 seed-2027 controls.

Llama3.2 shows the same broad pattern as Qwen3 in the matched control. Both payloads are fully detected in the native, INT8, INT4, mixing-0.75, and cache-retention conditions, while INT2 removes the signal for both methods. The clearest coding benefit appears under 50% projection pruning: BCH reaches 100% TPR at both offsets, whereas the uncoded control falls to 0% at offset 0 and 5% at offset 0.25. The pruning contrasts are 100 and 95 percentage points, with bootstrap intervals excluding zero. In contrast, coding does not provide a uniform advantage under weight mixing: at mix 0.50, the uncoded route is higher at offset 0, while the two methods are similar at offset 0.25. Thus, the second model confirms that the QIM control is informative mainly where pruning disrupts the carrier, not where the signal is already intact or fully erased.

The largest positive contrasts occur under projection pruning. At offset 0, BCH-minus-uncoded TPR is 60 percentage points at 30% pruning and 55 points at 50%, with paired intervals of [40,80] and [35,75]. At offset 0.25, the corresponding differences are 60 and 25 points, with intervals of [40,80] and [10,45]. The pruning rows show a coding benefit. The offset effect is not consistent: at 50% pruning, BCH TPR is lower at offset 0.25 than at offset 0.

Outside pruning, both payloads reach 100% under native, INT8, INT4, mixing 0.75, and visibility retention of 75% or 50%. Both fail under INT2 and complete mixing erasure. At 25% visibility, BCH remains at 100%, while the uncoded control reaches 90% at offset 0 and 85% at offset 0.25; both intervals include zero. In this control, the measurable coding advantage appears in the pruning rows; the other rows do not separate the two payloads.

4.3 Recovery, physical error, and prediction quality

Table 4 reports all lifecycle conditions for BCH interleaving. Panel (a) separates thresholded TPR from exact message recovery; panel (b) pairs payload BER with the change in natural-log perplexity.

Native and INT8 have zero BER and negligible prediction change. INT4 remains error-free on three checkpoints; Gemma3 reaches 76.7% TPR and 76.7% exact recovery. Under INT2, BER rises to 0.417, 0.526, 0.590, and 0.369 for Qwen3, Qwen2.5, Gemma3, and Llama3.2, respectively, and exact recovery is at most 6.7%.

Mixing separates message information from audit amplitude. At marked fraction 0.25, BER is zero and exact recovery is 100% on every checkpoint, but TPR is zero because correlation is below the fixed threshold. At fraction 0.5, the same zero-BER, 100%-exact messages yield checkpoint-dependent TPR at the strict boundary. Complete erasure produces BER 0.333 and zero exact recovery. Projection pruning changes both message recovery and acceptance: at 50% pruning, BER ranges from 0.019 on Llama3.2 to 0.412 on Gemma3, consistent with the broad TPR range in Table 1.

Visibility masking leaves TPR, exact recovery, and BER unchanged at 25% retained visibility, while Δ log PPL rises to 5.08, 5.96, 5.75, and 4.90. The layer-0 audit signal remains recoverable while next-token prediction quality degrades sharply.

Table 2: Complete QIM control for Qwen3, seed 2027. TPR and BER are percentages and fractions, respectively; $\Delta \log \mathrm{PPL}$ is relative to the native clean route. The interval is a paired 95% CI from document bootstrap resampling for BCH-minus-uncoded TPR, in percentage points. Offsets 0 and 0.25 are shown as separate subcolumns.

	Offset 0								Offset 0.25							
Transformation	BCH TPR	Unc. TPR	Diff.	95% CI	BCH	Unc.	BCH BER	Unc. BER	BCH TPR	Unc. TPR	Diff.	95% CI	BCH	Unc.	BCH BER	Unc. BER
Native	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.001	+0.000	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.002	+0.002
INT8	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.001	+0.001	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.002	+0.002
INT4	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.002	+0.002	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.004	+0.005
INT2	0.0	0.0	+0.0	[0,0]	0.421	0.450	+0.020	+0.022	0.0	0.0	+0.0	[0,0]	0.436	0.379	+0.028	+0.028
Mix 0.75	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.001	+0.001	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.002	+0.001
Mix 0.50	70.0	60.0	+10.0	[-10,30]	0.000	0.007	+0.001	+0.001	70.0	65.0	+5.0	[-10,20]	0.000	0.007	+0.001	+0.001
Mix 0.25	0.0	0.0	+0.0	[0,0]	0.000	0.086	+0.001	+0.001	0.0	0.0	+0.0	[0,0]	0.029	0.186	+0.001	+0.001
Mix 0	0.0	0.0	+0.0	[0,0]	0.636	0.657	+0.001	+0.001	0.0	0.0	+0.0	[0,0]	0.493	0.500	+0.000	+0.000
Prune 10%	100.0	95.0	+5.0	[0,15]	0.000	0.007	+0.000	-0.001	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.002	+0.000
Prune 30%	60.0	0.0	+60.0	[40,80]	0.214	0.407	+0.027	+0.025	60.0	0.0	+60.0	[40,80]	0.221	0.336	+0.030	+0.027
Prune 50%	55.0	0.0	+55.0	[35,75]	0.221	0.393	+0.059	+0.060	25.0	0.0	+25.0	[10,45]	0.329	0.379	+0.058	+0.059
Keep 75%	100.0	100.0	+0.0	[0,0]	0.000	0.000	+1.274	+1.274	100.0	100.0	+0.0	[0,0]	0.000	0.000	+1.272	+1.272
Keep 50%	100.0	100.0	+0.0	[0,0]	0.000	0.000	+3.286	+3.282	100.0	100.0	+0.0	[0,0]	0.000	0.000	+3.284	+3.281
Keep 25%	100.0	90.0	+10.0	[0,25]	0.000	0.014	+5.072	+5.077	100.0	85.0	+15.0	[0,30]	0.000	0.021	+5.079	+5.079

Table 3: Complete QIM control for Llama3.2, seed 2027, using the same configuration as the Qwen3 control. TPR and BER are percentages and fractions, respectively; $\Delta \log \mathrm{PPL}$ is relative to the native clean route. The interval is a paired 95% bootstrap CI for the BCH-minus-uncoded TPR difference, in percentage points.

	Offset 0								Offset 0.25							
Transformation	BCH TPR	Unc. TPR	Diff.	95% CI	BCH	Unc.	BCH BER	Unc. BER	BCH TPR	Unc. TPR	Diff.	95% CI	BCH	Unc.	BCH BER	Unc. BER
Native	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.003	+0.003	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.004	+0.004
INT8	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.003	+0.003	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.004	+0.004
INT4	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.005	+0.007	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.009	+0.010
INT2	0.0	0.0	+0.0	[0,0]	0.250	0.257	+0.086	+0.085	0.0	0.0	+0.0	[0,0]	0.264	0.271	+0.090	+0.091
Mix 0.75	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.003	+0.003	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.004	+0.003
Mix 0.50	40.0	70.0	-30.0	[-50,-10]	0.000	0.000	+0.003	+0.003	45.0	40.0	+5.0	[-10,20]	0.000	0.000	+0.001	+0.001
Mix 0.25	0.0	0.0	+0.0	[0,0]	0.000	0.036	+0.000	+0.001	0.0	0.0	+0.0	[0,0]	0.000	0.093	+0.000	-0.000
Mix 0	0.0	0.0	+0.0	[0,0]	0.636	0.643	+0.000	+0.000	0.0	0.0	+0.0	[0,0]	0.450	0.443	+0.000	+0.000
Prune 10%	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.003	+0.003	100.0	100.0	+0.0	[0,0]	0.000	0.000	+0.003	+0.004
Prune 30%	100.0	100.0	+0.0	[0,0]	0.000	0.000	-0.003	-0.003	100.0	100.0	+0.0	[0,0]	0.000	0.000	-0.006	-0.003
Prune 50%	100.0	0.0	+100.0	[100,100]	0.000	0.200	+0.027	+0.027	100.0	5.0	+95.0	[85,100]	0.000	0.186	+0.029	+0.025
Keep 75%	100.0	100.0	+0.0	[0,0]	0.000	0.000	+1.159	+1.155	100.0	100.0	+0.0	[0,0]	0.000	0.000	+1.159	+1.156
Keep 50%	100.0	100.0	+0.0	[0,0]	0.000	0.000	+3.072	+3.056	100.0	100.0	+0.0	[0,0]	0.000	0.000	+3.069	+3.053
Keep 25%	100.0	100.0	+0.0	[0,0]	0.000	0.000	+4.881	+4.884	100.0	100.0	+0.0	[0,0]	0.000	0.000	+4.901	+4.903

4.4 Payload construction across checkpoints

Table 5 gives the TPR matrix for interleaved BCH, ordered BCH, interleaved repetition, and the uncoded control. Each method uses the same number of physical coordinates and the same perturbation norm, so the table compares coding and symbol assignment.

Most benign cells give the same TPR. The visible differences occur in Gemma3 at INT4, in Qwen3 and Llama3.2 at INT2, at the mixing boundary of 0.5, and under projection pruning. Interleaved BCH is strongest on Qwen3 at 30% and 50% pruning and on Gemma3 at 30% pruning. There are also clear exceptions: at 50% pruning on Gemma3, repetition reaches 15% TPR versus 8.3% for interleaved BCH; on Llama3.2, ordered BCH reaches 100% versus 93.3%. At mixing 0.5, all methods recover the exact message and threshold rounding determines acceptance, so this row does not rank the payloads.

4.5 Fingerprint Persistence under Model Adaptation

To address our first research question, namely how much of the fingerprint survives model transformations, we evaluated fingerprinted checkpoints after multi-epoch downstream adaptation. We compared full-parameter SFT with LoRA SFT on English and Chinese training data, using matched clean checkpoints and the same held-out fingerprint probes at epochs 1, 3, 5, and 7.

Table 4: Complete BCH-interleaved recovery and quality matrix. Each cell in panel (a) is TPR/exact-recovery (percent/percent); each cell in panel (b) is payload BER/Δ log PPL. Values average three seeds and 20 documents per seed.

(a) Detection and exact message recovery

Transformation	Qwen3	Qwen2.5	Gemma3	Llama3.2
Native	100.0/100.0	100.0/100.0	100.0/100.0	100.0/100.0
INT8	100.0/100.0	100.0/100.0	100.0/100.0	100.0/100.0
INT4	100.0/100.0	100.0/100.0	76.7/76.7	100.0/100.0
INT2	5.0/6.7	0.0/0.0	0.0/0.0	0.0/1.7
Mix 0.75	100.0/100.0	100.0/100.0	100.0/100.0	100.0/100.0
Mix 0.50	61.7/100.0	28.3/100.0	30.0/100.0	28.3/100.0
Mix 0.25	0.0/100.0	0.0/100.0	0.0/100.0	0.0/100.0
Mix 0	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0
Prune 10%	100.0/100.0	100.0/100.0	100.0/100.0	100.0/100.0
Prune 30%	78.3/78.3	100.0/100.0	83.3/83.3	100.0/100.0
Prune 50%	43.3/43.3	93.3/93.3	8.3/8.3	93.3/96.7
Keep 75%	100.0/100.0	100.0/100.0	100.0/100.0	100.0/100.0
Keep 50%	100.0/100.0	100.0/100.0	100.0/100.0	100.0/100.0
Keep 25%	100.0/100.0	100.0/100.0	100.0/100.0	100.0/100.0

(b) Payload bit error and prediction loss

Transformation	Qwen3	Qwen2.5	Gemma3	Llama3.2
Native	0.000/+0.000	0.000/-0.000	0.000/+0.005	0.000/+0.002
INT8	0.000/+0.000	0.000/-0.000	0.000/+0.006	0.000/+0.002
INT4	0.000/+0.005	0.000/+0.000	0.076/+0.051	0.000/+0.009
INT2	0.417/+0.024	0.526/+0.041	0.590/+0.302	0.369/+0.083
Mix 0.75	0.000/+0.000	0.000/-0.000	0.000/+0.004	0.000/+0.002
Mix 0.50	0.000/+0.000	0.000/-0.000	0.000/+0.003	0.000/+0.001
Mix 0.25	0.000/+0.000	0.000/-0.000	0.000/+0.002	0.000/+0.000
Mix 0	0.333/+0.000	0.333/+0.000	0.333/+0.000	0.333/+0.000
Prune 10%	0.000/-0.000	0.000/-0.001	0.000/-0.004	0.000/+0.002
Prune 30%	0.107/+0.026	0.000/+0.001	0.050/+0.007	0.000/-0.002
Prune 50%	0.219/+0.056	0.031/+0.007	0.412/-0.028	0.019/+0.028
Keep 75%	0.000/+1.272	0.000/+1.739	0.000/+1.197	0.000/+1.156
Keep 50%	0.000/+3.289	0.000/+4.510	0.000/+3.036	0.000/+3.050
Keep 25%	0.000/+5.076	0.000/+5.963	0.000/+5.755	0.000/+4.897

As reported in Tables 6 and 7, fingerprint persistence is strongly model and training-condition dependent. Full-parameter updates preserve high detection rates for Qwen3 and Llama3.2, but cause a progressive decline for Qwen2.5 and, most notably, Gemma3 under Chinese adaptation. LoRA generally retains the fingerprint more effectively across the same lifecycle, indicating weaker disruption of the embedded carrier. The observed clean-model FPR remains negligible throughout. These results extend the transformation analysis beyond quantization, pruning, and cache operations, while also showing that pre-training detectability does not by itself guarantee persistence after downstream fine-tuning.

5 Conclusion

We introduce a coded, interleaved, spatially spread audit signal for Transformer KV caches. Across the tested transformations, BCH interleaving preserves exact audit decisions under native, INT8, most INT4, and cache-visibility retention. Its gain over uncoded payloads is concentrated under projection pruning; the matched QIM control measures gains of up to 60 percentage points in TPR. The experiments separate three outcomes: message recovery, decoder acceptance, and model utility. Zero payload BER can coexist with failed acceptance when correlation is low, and perfect recovery can coexist with a roughly five-unit increase in natural-log perplexity. INT2 quantization and aggressive pruning mark the main failure boundaries. The resulting protocol reports recovery, correlation, and quality separately and links redundancy to the transformation being tested.

6 Limitations and Future Work

Our experiments focus on a controlled cache-audit setting, using a set of checkpoints, seeds, transformations, and evaluation data. This scope supports the comparisons reported here and motivates three practical directions for future work. First, we will repeat the protocol on more checkpoints and model sizes to test how consistently the findings transfer. Second, we will evaluate real cache removal and longer-context serving, rather than controlled masking alone. Third, we will test whether the carrier remains detectable after common fine-tuning procedures. We will also add task-level evaluations alongside perplexity. These studies will extend the current results while preserving the core protocol.

7 Acknowledgments

Ethics and Privacy Statement. The experiments use publicly available pretrained language models and validation text; they have no human participants or private user data. The cache-audit mechanism supports model provenance and integrity checks. Hidden verification signals could also enable covert attribution, unauthorized tracking, or attempts to evade auditing. We report the decoder as audit evidence and evaluate it under controlled transformations.

Declaration of AI-assisted technologies. During the preparation of this work, the authors used ChatGPT (OpenAI) to provide limited assistance with drafting, modifying, and editing implementation code. After using this tool, the authors reviewed, tested, and verified all AI-assisted code and take full responsibility for the content of the publication.

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Table 5: Complete TPR comparison across payload constructions. Each cell reports BCH-I/BCH-O/Rep-I/Uncoded in percent, averaged over three seeds and 20 marked documents per seed. BCH-I, BCH-O, and Rep-I denote interleaved BCH, ordered BCH, and interleaved repetition.

Transformation	Qwen3	Qwen2.5	Gemma3	Llama3.2
Native	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0
INT8	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0
INT4	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	76.7/65.0/51.7/15.0	100.0/100.0/100.0/100.0
INT2	5.0/5.0/3.3/0.0	0.0/0.0/0.0/0.0	0.0/0.0/0.0/0.0	0.0/26.7/5.0/0.0
Mix 0.75	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0
Mix 0.50	61.7/35.0/38.3/65.0	28.3/75.0/40.0/55.0	30.0/6.7/8.3/38.3	28.3/63.3/65.0/56.7
Mix 0.25	0.0/0.0/0.0/0.0	0.0/0.0/0.0/0.0	0.0/0.0/0.0/0.0	0.0/0.0/0.0/0.0
Mix 0	0.0/0.0/0.0/0.0	0.0/0.0/0.0/0.0	0.0/0.0/0.0/0.0	0.0/0.0/0.0/0.0
Prune 10%	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0
Prune 30%	78.3/53.3/61.7/0.0	100.0/100.0/100.0/98.3	83.3/40.0/30.0/13.3	100.0/100.0/100.0/100.0
Prune 50%	43.3/0.0/0.0/0.0	93.3/36.7/35.0/30.0	8.3/6.7/15.0/1.7	93.3/100.0/76.7/26.7
Keep 75%	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0
Keep 50%	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0
Keep 25%	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0	100.0/100.0/100.0/100.0

Table 6: Full-parameter supervised fine-tuning (SFT) lifecycle table. E.k denotes epoch k.

Model	Lang.	E.1 TPR/FPR	E.3 TPR/FPR	E.5 TPR/FPR	E.7 TPR/FPR	E.7 BER	E.7 ΔPPL
Qwen2.5	EN	100/0	100/0	95/0	90/0	0.016	+0.0084
Qwen2.5	ZH	100/0	95/0	90/0	85/0	0.024	+0.0101
Qwen3	EN	100/0	100/0	100/0	100/0	0.000	+0.0068
Qwen3	ZH	100/0	100/0	100/0	95/0	0.008	+0.0087
Llama3.2	EN	100/0	100/0	100/0	100/0	0.000	+0.0142
Llama3.2	ZH	100/0	100/0	100/0	95/0	0.006	+0.0168
Gemma3	EN	95/0	90/0	80/0	70/0	0.052	-0.0048
Gemma3	ZH	90/0	75/0	60/0	40/0	0.118	-0.0031

Table 7: Low-rank adaptation (LoRA) SFT lifecycle table. This table follows the same checkpoints, languages, epochs, and audit metrics as Table 6.

Model	Lang.	E.1 TPR/FPR	E.3 TPR/FPR	E.5 TPR/FPR	E.7 TPR/FPR	E.7 BER	E.7 ΔPPL
Qwen2.5	EN	100/0	100/0	100/0	100/0	0.000	+0.0039
Qwen2.5	ZH	100/0	100/0	100/0	95/0	0.006	+0.0048
Qwen3	EN	100/0	100/0	100/0	100/0	0.000	+0.0032
Qwen3	ZH	100/0	100/0	100/0	100/0	0.000	+0.0041
Llama3.2	EN	100/0	100/0	100/0	100/0	0.000	+0.0067
Llama3.2	ZH	100/0	100/0	100/0	100/0	0.000	+0.0075
Gemma3	EN	100/0	100/0	95/0	90/0	0.015	-0.0025
Gemma3	ZH	100/0	95/0	90/0	80/0	0.031	-0.0018

IEEE Transactions on Neural Networks and Learning Systems 35, 10 (2024), 13082–13100. doi:10.1109/TNNLS.2023.3270135

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